

Which is Better?



Prompt A

Prompt B



@akshay_pachaar

So, how prompt selection is done ?

An LLM is tested on a certain test dataset using different prompt templates & the performance is evaluated.

The challenge with real world data is that it can be noisy.

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Clean labels are not always available
& it can be very costly to create them!

Text Example

Are you crazy down here?! What the heck is going on?

Thanks for your response! How should we proceed ?

Noisy Label

polite

impolite

Collected from a
single annotator

Clean Label

impolite

polite

Consensus of
multiple annotators

Today, there are various ways to prompt LLMs.

Here are a few standard techniques:

- Few Shot prompting
- Instruction prompting
- Templatized prompting
- Chain of thoughts prompting

Let's understand them one by one!

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Chain of Thought

Prompt A

Question. Employing your vast language proficiency as an exceptional classifier, could you assess the sentiment conveyed in this text?: 'I didn't see your internal link, I put it back. It isn't orphaned, what are you talking about?'

Answer: Negative. The answer is negative because 'what are you talking about' is passive aggressive and rude.

Question. Employing your vast language proficiency as an exceptional classifier, could you assess the sentiment conveyed in this text?: Greetings, and thank you for your cogent remarks at. Would you be interested in voting in the straw poll as well?

Answer: Positive. The answer is positive because 'greetings' and 'thank you' are nice things to say.

Question. Employing your vast language proficiency as an exceptional classifier, could you assess the sentiment conveyed in this text?: {text sample}

Answer:

Prompt B

Question. What is the sentiment of the following text: 'I didn't see your internal link, I put it back. It isn't orphaned, what are you talking about?'

Answer: Negative. The answer is negative because 'what are you talking about' is passive aggressive and rude.

Question. What is the sentiment of the following text: Greetings, and thank you for your cogent remarks at. Would you be interested in voting in the straw poll as well?

Answer: Positive. The answer is positive because 'greetings' and 'thank you' are nice things to say.

Question. What is the sentiment of the following text: {text sample}

Answer:

Instruction

Prompt A

Please classify the sentiment of the following text as positive or negative.

Text: {text sample}
Sentiment:

Prompt B

What is the sentiment of this text? Choose from 'positive' or 'negative'.

Text: {text sample}
Sentiment:

Few-Shot

Prompt A

Here are examples of texts and their sentiments.

I didn't see your internal link, I put it back. It isn't orphaned, what are you talking about?: Negative
Greetings, and thank you for your cogent remarks at . Would you be interested in voting in the straw poll as well?: Positive

Please classify the sentiment of the following text as positive or negative.

{text sample}:

Prompt B

I'm giving you examples of texts and their sentiments.

I didn't see your internal link, I put it back. It isn't orphaned, what are you talking about?: Negative
Greetings, and thank you for your cogent remarks at . Would you be interested in voting in the straw poll as well?: Positive

Determine the sentiment of the following text. Choose from 'positive' or 'negative'.

{text sample}:

Templatized

Prompt A

What is the sentiment of the following text?

I am giving you the text and two choices. 'A' for 'positive' and 'B' for 'negative'.

{text sample}

A. Positive

B. Negative

Prompt B

What is the sentiment of this text?

{text sample}

A. Positive

B. Negative

Ok, Let's evaluate some prompts, we have:

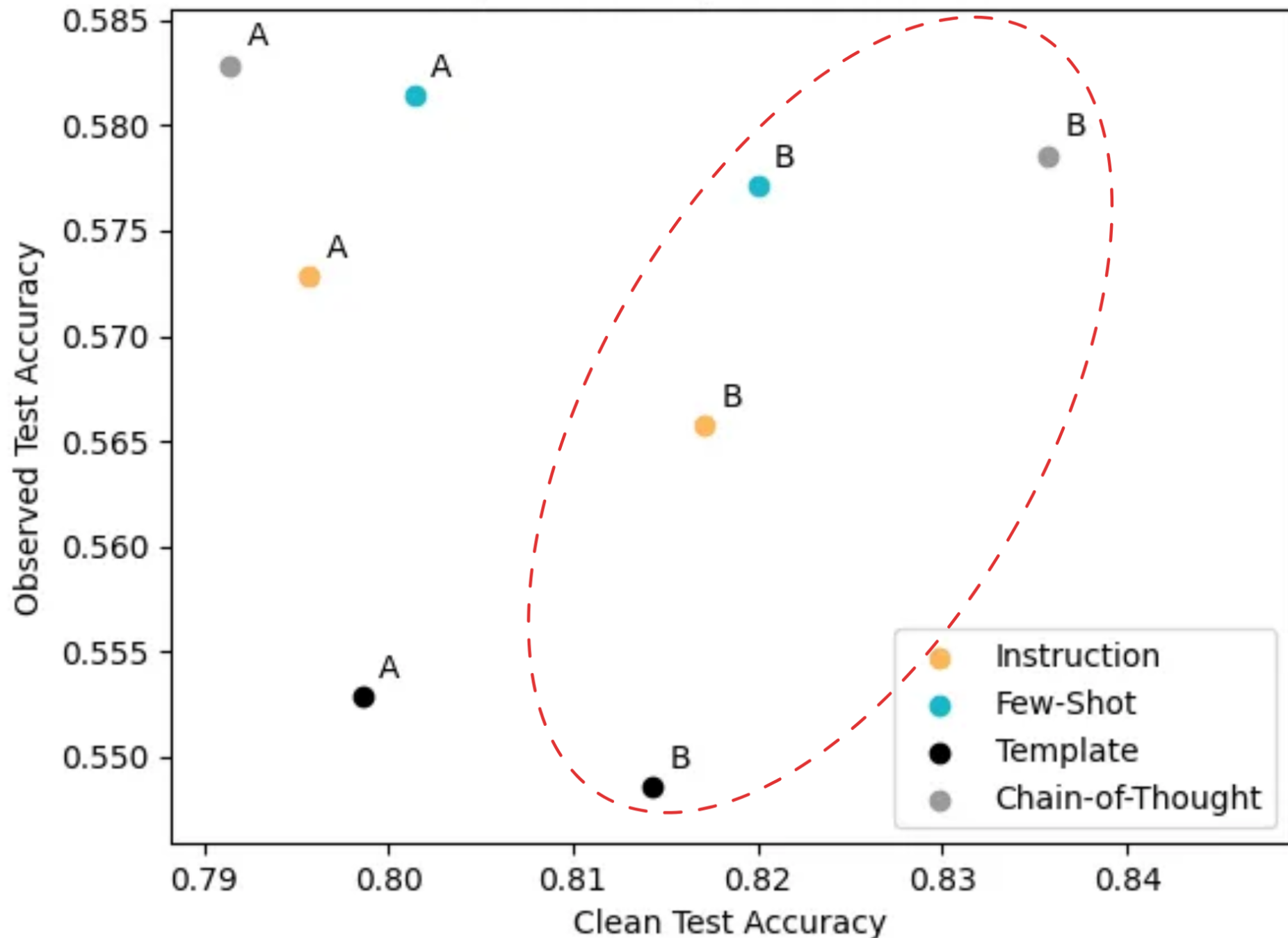
- Model: FLAN-T5 LLM
- Dataset: Stanford Politeness
- Prompts: Two prompts A & B from each of the 4 standard prompting techniques.

Evaluation is done on both noisy (observed) & clean data, to see the impact of noisy labels.

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Test Accuracy on clean data appears to be much better for prompt B in each of the 4 settings!

Impact of Noisy Evaluation Data



Without clean data for evaluation, we may choose the wrong prompt.

There are two ways to get clean data:

1. Manual correction, very costly ❌
2. Algorithmic Correction ✅

So let's talk about algorithmic correction!

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Notice how data can affect the choice of prompt!

	observed accuracy	clean accuracy
● Instruction ● Few-Shot ● Template ● Chain-of-Thought Prompt A	58.3%	79.1%
Prompt B	57.8%	83.6%
Prompt A	58.1%	80.1%
Prompt B	57.7%	82.0%
Prompt A	57.3%	79.5%
Prompt B	56.5%	81.7%
Prompt A	55.2%	79.8%
Prompt B	54.8%	81.4%

Researchers at MIT have developed Confident Learning algorithms that automatically clean any dataset's labels!

After applying these algos to our original data, prompt evaluation becomes more reliable.

You can now correctly identify which prompt is better!

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 @akshay_pachaar

Evaluation done using:

- Instruction
- Few-Shot
- Template
- Chain-of-Thought

Raw Data

Cleaned using
Confident Learning

Manually
Cleaned

observed
acc

CL
acc

clean
acc

Prompt A

58.3%

72.8%

79.1%

Prompt B

57.8%

76.4%

83.6%

Prompt A

58.1%

74.1%

80.1%

Prompt B

57.7%

76.0%

82.0%

Prompt A

57.3%

73.6%

79.5%

Prompt B

56.5%

76.0%

81.7%

Prompt A

55.2%

73.8%

79.8%

Prompt B

54.8%

74.5%

81.4%

Notice that the results using Confident Learning & Manually Cleaned data are aligned & it happens at zero cost!!

That's a wrap!

If you interested in:

- Python 
- Data Science 
- Machine Learning 
- MLOps 
- NLP 
- Computer Vision 
- LLMs 

Follow me on LinkedIn 

Everyday, I share tutorials on above topics!

Cheers!! 



@akshay_pachaar